

M1 Internship Report - Bioinformatics & Biological Platform Engineering

Identification of driver genes for drug response prediction in lung adenocarcinoma using multi-omic data and machine learning.

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Introduction

According to the World Health Organization (WHO), cancer remains the second leading cause of death worldwide, with lung cancer being the leading cause of cancer-related deaths. Lung adenocarcinoma (LUAD) is the most common subtype of non-small cell lung cancer (NSCLC), accounting for approximately 85% of cases. Although smoking is a major risk factor, the emergence of cases among nonsmokers highlights the involvement of additional risk factors, including genetic predisposition [1, 2]. One of the goals of precision oncology is to identify the best treatment options to spare patients the severe side effects of inappropriate ones. [3]

The literature demonstrates that the integration of multi-omics data outperforms single-omics approaches in predicting drug sensitivity and identifying disease driver genes [4]. Recent benchmarking studies show that while statistical differences between various deep intermediate integration methods (projecting each omics layer into a latent space before merging these compressed representations) themselves are rarely observable, early integration (raw concatenation) performs significantly worse due to feature sparsity and the curse of dimensionality [5]. Furthermore, integrating the chemical structure of drugs and leveraging protein-protein interaction (PPI) networks or metabolic pathways makes it possible to better model the cellular response and to generalize predictions to novel drugs [6].

However, current architectures face several limitations. First, the models suffer from the curse of dimensionality (more variables than observations) [7]. Second, most models predict therapeutic response on a pan-cancer scale, obscuring tissue-specific resistance mechanisms [8]. Third, the use of various deep learning methods within a single model enables very good performance but drastically reduces interpretability, which impacts their clinical application. Finally, when models trained purely on in vitro data (cancer cells cultured in the laboratory) are tested on in vivo data, the predictions fail due to the biological gap [7].

The objective of this study is, first, to identify candidate driver genes for LUAD so that their multi-omic profiles can be used as predictors of the drug response. The first phase has been completed. The second phase has begun and will be carried out by another intern. It is important to note that the project recognizes that disease driver genes are not necessarily targeted by drug treatments.

Materials and Methods

Candidate Driver Genes Identification

All Phase 1 analyses were performed using R version 4.5.0, with the exception of the GISTIC 2.0 tool which required a Linux Ubuntu environment.

Data retrieval

The objective of this phase was to identify driver genes for LUAD. To this end, clinical and omics data (expression, mutations, copy number alterations) were collected from the TCGA-LUAD project (The Cancer Genome Atlas – Lung Adenocarcinoma) on the public Genomic Data Commons (GDC) portal. [9]

For the transcriptomic profile, gene expression quantification data (generated by *the STAR-Counts* [10] alignment algorithm) were extracted, totaling 601 files for 518 patients.

For the somatic mutations, the data analyzed were the SNV-type (single nucleotide variant) masked somatic mutations generated using the *Aliquot ensemble somatic variant merging and masking* [11] pipeline. The initial download included 618 files (559 patients), but during merging using the *maftools* package, two files were automatically excluded due to a complete absence of mutations, reducing the final cohort to 616 files.

For the copy number alterations, gene-level estimates and allele segments (*Affymetrix SNP 6.0* platform) were retrieved. These data are derived from the *ASCAT3* [12] (Allele-Specific Copy Number Analysis of Tumors) workflow. For each of these two categories, 503 files corresponding to 503 patients were downloaded.

The OncoKB [13] and COSMIC CGC [14] databases were queried and provided respectively 33 and 46 recognized driver genes (required for the machine learning stage).

To explore possible transcriptional regulatory mechanisms governing the candidate genes identified by machine learning models, ChIP-seq (chromatin immunoprecipitation coupled with sequencing) data were analyzed focusing on FOXA1, a transcription factor frequently dysregulated in epithelial cancers, by examining data from the A549 cell line, a model for human lung adenocarcinoma. Although other major regulators of LUAD (such as FOXJ1, GATA6, MYCN, and NKX2-1) were searched in the ENCODE database [15], no binding data were available for this specific cell line.

Gene expression

The transcriptomic unstranded counts files were merged into a matrix combining the 601 samples. To optimize the statistical power of the analysis, pre-filtering was performed using the *filterByExpr* function from the *edgeR* package to eliminate genes with low expression levels in both healthy and tumor tissues. Differential expression analysis, which determines whether gene expression differs significantly between two conditions (here between healthy and tumor tissues), was then performed using the *DESeq2* package [16]. The statistical model was designed to isolate the effect of the tumor by including smoking status and gender as clinical covariates ($\text{design} = \sim \text{smoking_status} + \text{gender} + \text{condition}$). To limit the number of false positives generated by the simultaneous analysis of thousands of genes, p-values were adjusted using the Benjamini-Hochberg method [17]. Differentially

expressed genes (DEGs) were defined using a double threshold where the adjusted pvalue < 0.01 and the absolute value of $\log_2$ Fold Change > 1 .

Copy number alteration - CNA

To assess genomic instability at the cohort level, the segmentation data were processed using the *GISTIC 2.0* algorithm [18]. The tool was run using the Docker container developed and maintained by Dr Shixiang Wang [19]. The confidence level was set to 0.99 (-conf 0.99) to estimate the spatial boundaries of the resulting alteration peaks. To minimize false positives, only chromosomal regions identified with strict significance (q-value < 0.1) were selected to generate the list of genes considered significantly amplified or deleted for integration purposes. At the same time, a matrix compiling the absolute copy number per gene and per patient was constructed, underwent quality control and was annotated with the dominant event (Amplification, Deletion, Neutral).

Mutations

Mutation files were aggregated and processed using the *maftools* package. To identify genes under positive evolutionary selection, the *dN/dS* algorithm [20] was run using the *dndscv* package. The algorithm estimated the selection ratios for missense, nonsense, and splice site mutations. The statistical significance of these different types of alterations was combined and adjusted using the Benjamini-Hochberg method to obtain an overall q-value (qglobal_cv), which quantifies the probability that the gene is under positive selection pressure, meaning that the accumulation of its alterations confers an advantage (in terms of proliferation or survival) to the tumor cell, indicating its potential role as a driver gene. Only genes that simultaneously had a qglobal_cv < 0.01 and at least one selection ratio strictly greater than 1 (w_mis , w_non , or $w_spl > 1$) were kept for the rest of the study.

Integration and Pathways Analysis

A matrix, combining all the genes and the results of these independent omics analyses, was created. First, a strict selection (Tier 1) identified genes that were significantly altered in all three layers simultaneously (**expression**: $p_adj < 0.001$ and $|\log_2 FC| > 2$, **CNA GISTIC**: $q < 0.1$, **Mutation**: $q < 0.01$ and selection ratio > 1). Then, a more relaxed selection (Tier 2) retained genes altered in at least two of these three layers. Tier 2 will be studied by the machine learning models.

In parallel, a gene set enrichment analysis (GSEA [21]) was conducted to identify metabolic pathways dysregulated by LUAD. The entire transcriptome was ranked in descending order according to the Wald statistic (ratio of the $\log_2FC$ to its standard error) calculated by *DESeq2*. This metric allows genes to be ranked from the most significantly overexpressed to the most underexpressed. Enrichments were calculated using the Gene Ontology (function *gseGO*, ont = 'BP') and KEGG (function *gseKEGG*, organism = 'hsa') reference databases. The core enrichment genes, which drive

the dysregulation signal in these pathways, were extracted and incorporated into the multi-omic matrix.

Prediction

To train the prediction models, 33 LUAD driver genes from OncoKB and the 46 NSCLC driver genes from the COSMIC database were merged and deduplicated, creating a reference list of 65 genes. Each gene was characterized by ten continuous and Boolean predictive variables derived from previous statistical tests: log2FC, padj, mut_freq_norm, mut_q, Pct_Amplified, Pct_HomDel, in_CNA_gistic, in_CNA_cnv_threshold, in_GO_pathway, and in_KEGG_pathway.

The final dataset (979 genes) was constructed by combining these 65 drivers, the Tier 2 candidate genes (exhibiting at least two significant omic alterations), and a random sample of genes considered neutral (class imbalance of 14.08 non-driver to 1 driver). This dataset was split, using stratified partitioning (*caret* package), into a training set (80%, 784 genes) and a test set (20%, 195 genes).

The evaluation of the models was based on repeated cross-validation (5 folds, 5 repetitions for a total of 25 resamplings) measuring a set of performance metrics. Among these, the area under the ROC curve (AUC) quantified the overall discriminatory power, while the F1 score (corresponding to the harmonic mean of precision and sensitivity) evaluated the robustness of performance given class imbalance. Sensitivity was used to determine the ability to identify true driver genes, while specificity assessed the ability to distinguish non-driver genes. Finally, the Matthews correlation coefficient (MCC), which considers the entire confusion matrix, allows for the evaluation of a model's true predictive power when facing an unbalanced sample.

A grid search was conducted to optimize the hyperparameters, and class weighting was incorporated to compensate for data imbalance. The Random Forest model [22], consisting of a set of decision trees combined via majority voting, was implemented using the *caret* package (rf method) and optimized for the F1 score. Similarly, the Elastic Net [23] logistic regression (glmnet method, same package) was optimized for the F1 score by combining the L1 penalty (Lasso which sets the coefficients of non-informative variables to zero) and the L2 penalty (Ridge which reduces the coefficients of variables close to zero). Finally, the XGBoost [24] (Extreme Gradient Boosting) model, which builds trees sequentially to iteratively correct the errors of the previous one, was trained using the *xgboost* package and was optimized for AUC.

To explain the predictions, a SHAP [25] (SHapley Additive exPlanations, *SHAPforxgboost* package) analysis was conducted on the XGBoost model. It quantifies the overall impact of each variable on the model's predictions.

Regulation (Transcription factor)

To investigate the hypothesis of regulation by the FOXA1 transcription factor, the candidate genes were cross-referenced with chromatin immunoprecipitation (ChIP-seq) data from the A549 lung cell line. The enrichment peaks were annotated and only genes showing a peak of direct FOXA1 binding their promoter were extracted. These physical targets were then intersected with the top 10 ‘novel’ candidates from Random Forest and XGBoost to identify possible double activation events (genes targeted by the transcription factor and exhibiting chromosomal amplification). Finally, to support this hypothesis of potential regulation, the transcriptomic expression levels between FOXA1 and these candidate genes were evaluated using the Spearman correlation coefficient.

Drug Response Prediction

The objective of this second phase was to develop a predictive model to estimate drug response, quantified by the median Inhibitory Concentration (IC50), in in-vitro cell lines. Long term, the protocol aimed to apply the model, via direct inference (zero-shot), to in-vivo data from patients of the TCGA-LUAD cohort. The Phase 2 analyses were performed in Python (version 3) using the Google Colaboratory environment.

Data Retrieval

Drug sensitivity data (GDSC1 and GDSC2 cohorts), including the target variable $\ln(\text{IC}_{50})$, were extracted from the Genomics of Drug Sensitivity in Cancer project on the Cell Model Passports portal [26]. The molecular representations of each compound, in SMILES format, were added to the dataset by querying the PubChem [27] and ChEMBL [28] databases.

To characterize these cell lines, the corresponding multi-omic profiles were obtained from the DepMap portal [29] (version 25Q2). The data include the expression of coding genes ($\log_2(\text{TPM}+1)$, where TPM stands for Transcripts Per Million), copy number alterations ($\log_2(\text{CN}+1)$, where CN stands for Copy Number), and somatic mutations. The raw file, listing mutational events, was binarized to indicate the presence (1) or absence (0) of a mutation for each gene within the cell lines. For the patient inference step (Zero-Shot Inference), the TCGA-LUAD data acquired during Phase 1 were reused. These data were harmonized to match the training data format.

To optimize dimensionality, the features were restricted to the candidate genes (Tier 2) having a probability $\geq 60\%$ of being a driver (XGBoost model), the 33 clinically validated driver genes from the OncoKB database and pharmacokinetic genes (ADME: Absorption, Distribution, Metabolism, and Excretion) extracted from the ClinPGx database [30].

Preprocessing & Cold-Cell Split

Data preparation began with the alignment of expression, mutation, and copy number alteration matrices, which were then linked to the GDSC drug response database. Only cell lines with complete data for all three omics and drug response data were retained (504 cell lines).

To prevent data leakage, the datasets were split using the “Cold-Cell Line Splitting” approach. The splitting was performed at the level of the entire cell line, rather than by cell line-drug pair. Grouping by cell line ensures that the same cell line does not appear simultaneously in both the training and validation sets. Thus, 80% of the cell lines (403) were assigned to the training set and 20% (101) to the test set. For hyperparameter optimization, 5-fold cross-validation (using the *GroupKFold* function from the *sklearn* package) was performed on the training set.

Drug Encoder

First, the molecules were cleaned (salts and solvents removed) using the *RDKit* library. Then, SMILES strings were standardized so that each molecule would have a single and unique computational representation. Each molecule was converted into a graph using the *PyTorch* Geometric library. A graph neural network based on the AttentiveFP architecture [31] was used to process these molecules (message passing and graph pooling). It transforms each molecular graph into a 256-dimensional vector (Zdrug). This vector is to be later combined with the latent representations of cellular omics data to feed the model of drug response prediction.

Results

Individual analysis of the omics layers validated the quality and processing of our data. Differential expression analysis identified significantly dysregulated genes (*DESeq2*, adjusted p-value < 0.01). As shown in the Volcano Plot (**Appendix 1**), the transcriptome reveals the expected overexpression of cancer-related genes, such as FAM83A. The analysis of the mutational landscape confirms the predominance of TP53 and KRAS alterations (**Appendices 2**) as well as the C>A transversion signature associated with smoking. Finally, analysis of copy number alterations using *GISTIC* identified chromosomal regions that were significantly amplified or deleted (default *GISTIC* threshold $q < 0.25$, **Appendix 3**), from which a stricter subset ($q < 0.1$) was retained for downstream gene-level analysis. The joint analysis of expression, copy number, and mutations identified genomic alterations considered significant. The intersection of these three layers (Tier 1) proved too restrictive, retained only a single gene (SPRR1B). A more flexible approach requiring significance in at least two out of three layers (Tier 2) generated a pool of 593 candidate genes. In parallel, GSEA revealed an activation of cell cycle, DNA replication, and ribosome pathways, alongside a suppression of

immune and vascular pathways (angiogenesis regulation, blood circulation) consistent with the proliferative phenotype of LUAD (**Appendix 4**).

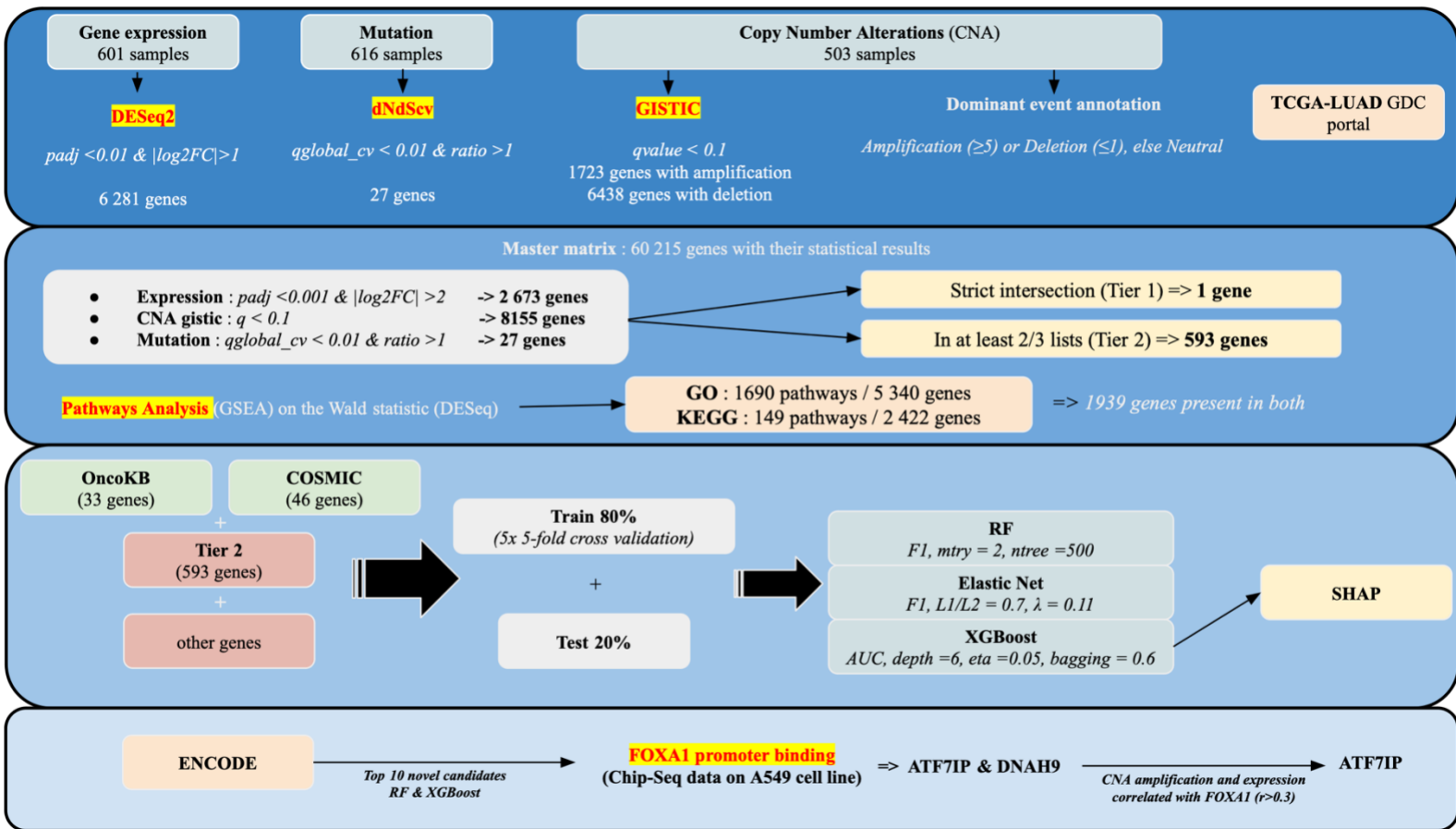


Figure 1: Overview of the Phase 1 workflow

Each panel corresponds to an analytical stage, read from top to bottom. **First panel:** the three omics layers, retrieved from the TCGA-LUAD project on the GDC portal, were processed independently (DESeq2, dNdScv, and GISTIC, respectively), each yielding its own list of significant genes. **Second panel:** the three gene lists are merged into a master matrix, from which two integration strategies are derived. In parallel, pathway enrichment analysis identifies additional dysregulated gene sets. **Third panel:** Tier 2 candidates are combined with reference driver genes (OncoKB, COSMIC) and split into training and test sets to train three machine learning models. Predictions were interpreted using SHAP. **Fourth panel:** ChIP-seq data from the ENCODE database were used to test whether the top novel candidate genes are physically targeted by the FOXA1 transcription factor, and whether this binding co-occurs with genomic amplification and a positive expression correlation with the transcription factor.

As shown in **Figure 2**, genes from FAM107A to ACVRL1 are underexpressed while genes from CRAMP2 to UHRF1 are overexpressed in tumor tissues. Transcriptomic expression within tumor samples remains heterogeneous both between samples and between genes. This heatmap serves as a quality control step confirming that DESeq2 separates correctly tumor from healthy samples, with known oncogenes, such as FAM83A, recovered among the top overexpressed genes.

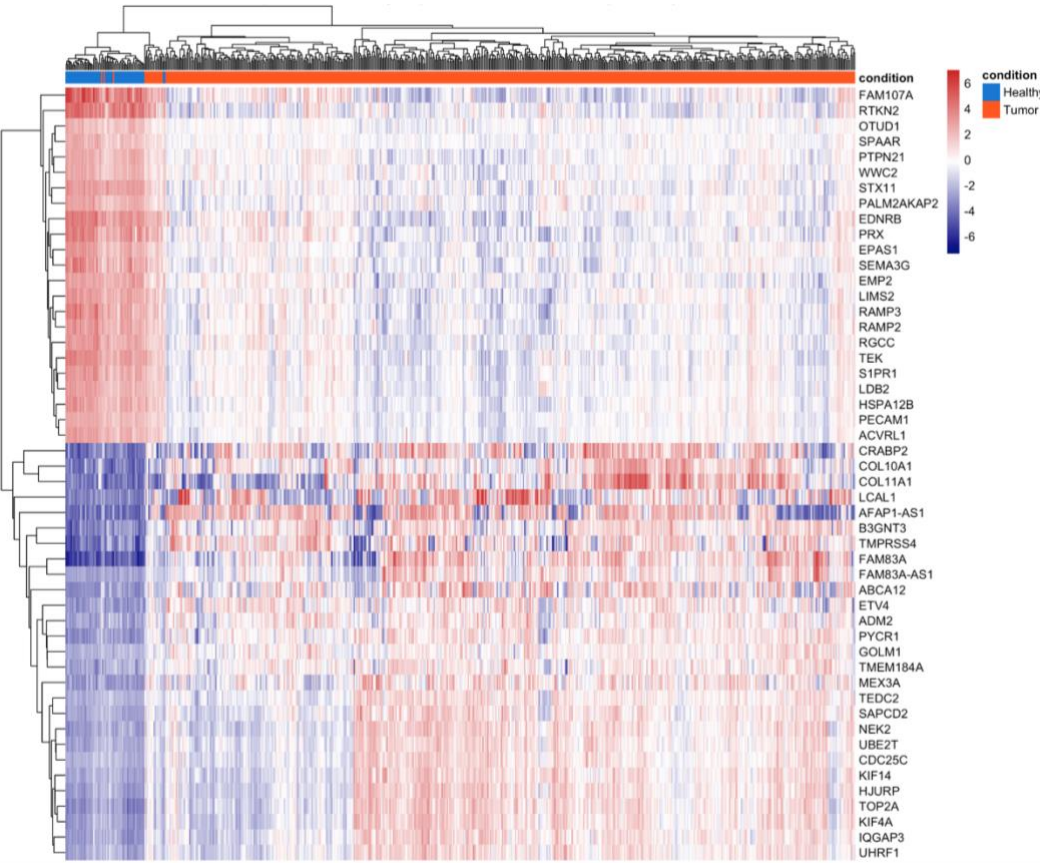


Figure 2: Heatmap of the 50 most DEGs of the TCGA-LUAD

The rows represent genes and the columns correspond to patient samples (healthy tissue in blue and tumor tissue in orange). The color scale illustrates expression levels, ranging from blue (underexpression) to red (overexpression). Genes and samples are clustered hierarchically based on the similarity of their expression profiles.

A)

Model	ROC	Accuracy	BalancedAcc	Sensitivity	Specificity	F1	MCC
Random Forest	0.950 ± 0.013	0.933 ± 0.014	0.716 ± 0.079	0.465 ± 0.162	0.966 ± 0.012	0.469 ± 0.135	0.440 ± 0.139
Elastic Net	0.832 ± 0.069	0.878 ± 0.028	0.801 ± 0.064	0.714 ± 0.132	0.889 ± 0.030	0.439 ± 0.079	0.423 ± 0.089
XGBoost	0.932 ± 0.034	0.916 ± 0.021	0.845 ± 0.064	0.764 ± 0.128	0.927 ± 0.021	0.549 ± 0.092	0.534 ± 0.100

B)

Model	ROC	Accuracy	BalancedAcc	Sensitivity	Specificity	F1	MCC
Random Forest	0.971	0.954	0.797	0.615	0.978	0.640	0.616
Elastic Net	0.919	0.892	0.799	0.692	0.907	0.462	0.439
XGBoost	0.973	0.944	0.934	0.923	0.945	0.686	0.684

Table 1: Performance of the Random Forest, Elastic Net, and XGBoost models for classifying driver genes of TCGA-LUAD. Results for the cross-validation (A) and the test set (B)

Cross-validation performances are expressed as the mean ± standard deviation across 25 resampling runs (5-fold cross-validation repeated 5 times) on the training set.

This analysis (Table 1) highlights the advantage of nonlinear approaches in dealing with the complexity of multi-omic data. Elastic Net linear model exhibits more modest cross-validation performances (A) (ROC of 0.832±0.069), whereas Random Forest achieves an ROC of 0.950±0.013 as well as a high specificity (0.966±0.012), but still suffers from low sensitivity (0.465±0.162). XGBoost deals better with class imbalance and is the best performing model achieving the highest

balanced accuracy (0.845 ± 0.064), the best sensitivity (0.764 ± 0.128), as well as the best F1 scores (0.549 ± 0.092) and MCC (0.534 ± 0.100).

Evaluation on the independent test set (**B**) confirms the robustness of XGBoost. This architecture outperforms all the others across all balanced metrics with a ROC of 0.973, a balanced accuracy of 0.934, a sensitivity of 0.923, a F1 score of 0.686, and a MCC of 0.684. XGBoost reliably classifies true driver genes while minimizing false negatives.

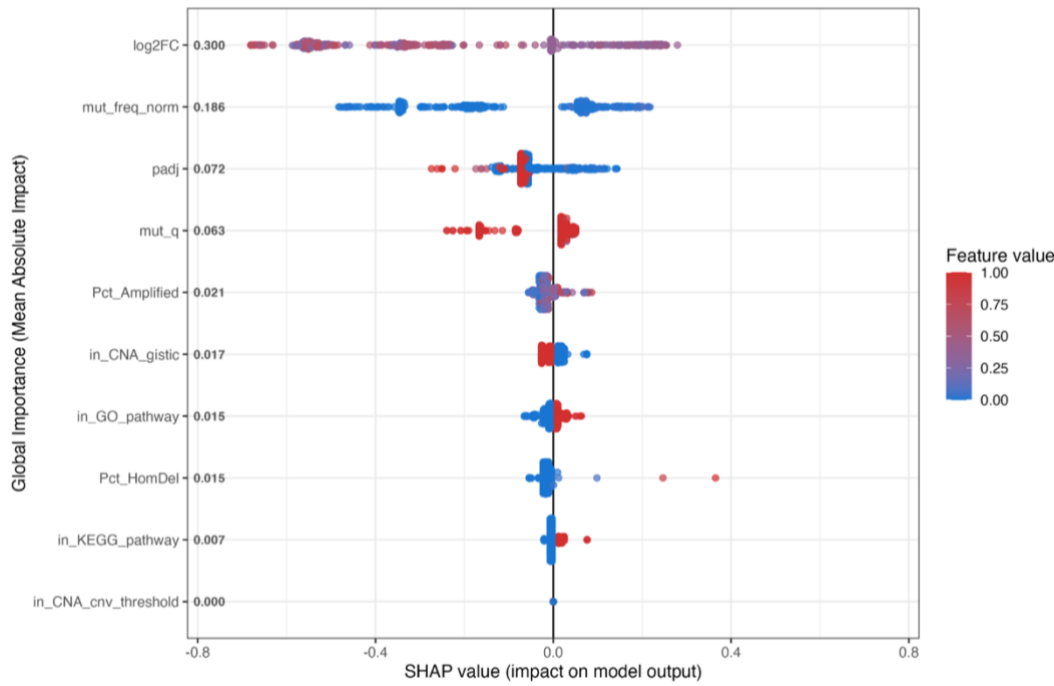


Figure 3: Global impact distribution of the variables on XGBoost predictions

The 10 variables were ranked in descending order of importance (mean absolute value of the impact is displayed next to each variable name). The SHAP value is plotted on the x-axis. A positive score increases the probability that the model will classify the gene as a driver, while a negative score favors the non-driver class. Each point represents a gene from the training set. The color scale illustrates the value of the variable for that gene, ranging from blue (low value or False) to red (high value or True).

Two variables dominate the predictions (**Figure 3**). The log2FC (magnitude of transcriptional dysregulation) and the normalized mutation frequency account for almost half of the model's total explanatory power. The other variables contribute to a smaller extent. A small value of the adjusted pvalue (padj), meaning that the transcriptional dysregulation is significant, pushes the prediction to the driver class as well as a high percentage of homologous deletion (Pct_HomDel). If a gene was identified in either GO or KEGG pathways, the model tends to classify it as a driver.

XGB_probability	XGB_genes	RF_genes	RF_probability
0.64	KRAS	EGFR	0.944
0.64	ATF7IP*	KEAP1	0.936
0.64	ARID1A	STK11	0.882
0.64	ATM*	KRAS	0.858
0.63	STK11	SMARCA4	0.856
0.63	TP53	ARID1A	0.816
0.63	SMARCA4	TP53	0.764
0.63	MGA*	RAD51	0.67
0.63	SETD2*	CDKN2A	0.628
0.63	KEAP1	ATM*	0.614

Table 2: Top 10 genes predicted by the XGBoost (XGB) and Random Forest (RF) models.

Genes in bold appear in the top rankings of both algorithms, and those marked with an asterisk (*) represent the 'novel' candidates not found in the reference databases (OncoKB, COSMIC).

Table 2 shows strong consistency in the identification of known LUAD driver genes such as KRAS and TP53 identified by both algorithms. However, the probabilities are quite distinct due to architectural differences between the models. RF assigns higher and

more widely distributed probabilities (for example 85.8% for KRAS) while XGBoost presents narrower probabilities ranging from 63% to 64%. Finally, XGBoost identified four ‘novel’ candidates (ATF7IP, ATM, MGA, and SETD2). Among these, the ATM gene also ranks in the top 10 of the RF model (with a probability of 61.4%).

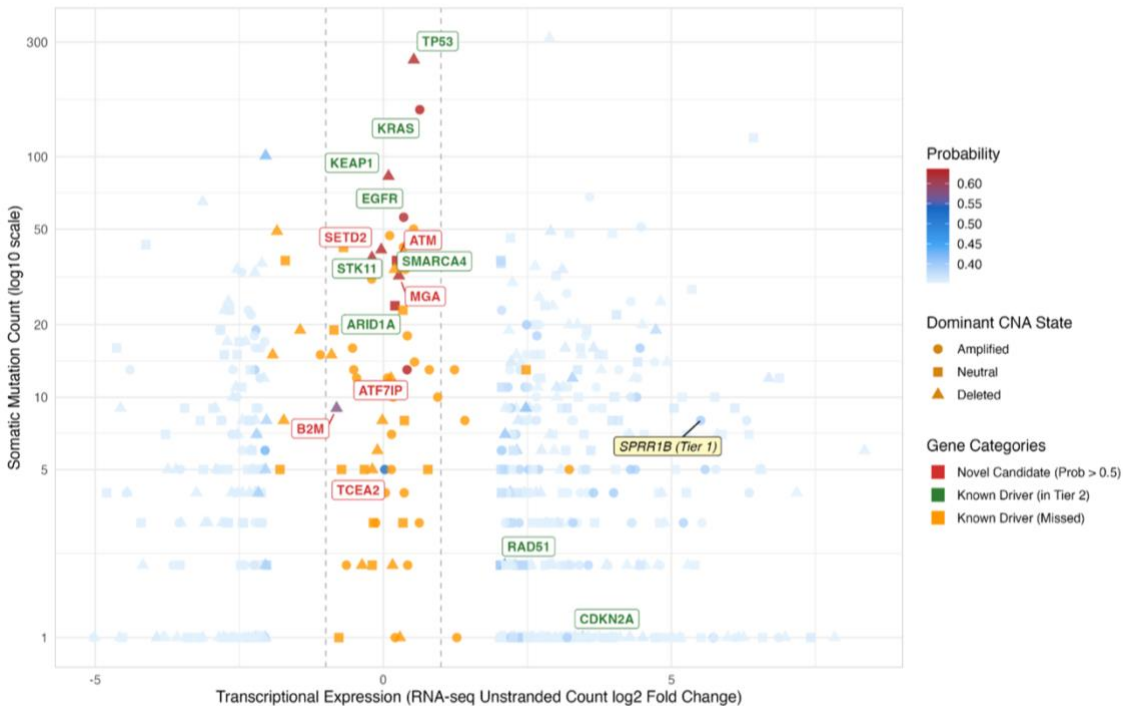


Figure 4: Multi-Omics Landscape of TCGA-LUAD

The x-axis represents the differential gene expression ($\log_2FC$). The y-axis shows the number of somatic mutations per gene on a logarithmic scale. Each point corresponds to a gene, with its color representing the probability assigned by the XGBoost model (from blue for a low probability to red for a high probability of being a driver gene). The shapes indicate the status of copy number alterations. Finally, the labels identify driver genes in Tier 2 (in

green), true driver genes excluded from Tier 2 (in orange), ‘novel’ candidates (in red), as well as the single gene resulting from the strict intersection of Tier 1 (SPRR1B, in yellow).

The multi-omics landscape (**Figure 4**) highlights the limitations of a strict intersection approach. Although SPRR1B is the only gene to be statistically significant across all three omics layers (Tier 1), XGBoost assigns it a very low probability of being a driver. In contrast, the genes in Tier 2 are characterized by strong signals across at least two dimensions (for example, a high mutational burden combined with copy number alterations). This reflects the very design of Tier 2, which required statistical significance in at least two of the three omics. However, this rigid constraint led to the exclusion of many true driver genes (shown in orange), either because they did not meet the significance thresholds or because they exhibited major alterations in only a single omic dimension. The ‘novel’ candidate genes identified by XGBoost (SETD2, ATM, and MGA) exhibit a multi-omic profile (spatial location and copy number alteration status) similar to true driver genes such as STK11, SMARCA4, and ARID1A, consistent with the high predictive probability assigned to them by the model. In addition, the analysis of the transcription factor FOXA1 (**Figure 1**, fourth panel) reveals that both ‘novel’ candidate genes ATF7IP and DNAH9 are physically bound by FOXA1 to their promoter. Only ATF7IP also displays a genomic amplification and a positive expression correlation Spearman ($r > 0.4$) with this regulator.

Discussion

The aim of this study was to identify a panel of potential driver genes for lung adenocarcinoma in order to compile their multi-omic profile and, ultimately, predict patients drug response.

The results demonstrate that a strict intersection approach between the transcriptome, copy-number alterations, and somatic mutations (Tier 1) is too conservative and unsuitable for tumor complexity, identifying only a single gene (SPRR1B). In contrast, adopting a more flexible strategy (Tier 2, 593 genes), combined with supervised machine learning, overcomes this limitation.

The XGBoost model effectively identifies true driver genes of lung adenocarcinoma (KRAS, TP53, ARID1A) while also identifying ‘novel’ candidates absent from reference databases (ATF7IP, ATM, MGA, SETD2). The SHAP analysis shows that these predictions are primarily based on the magnitude of transcriptomic dysregulation (log2FC) and normalized mutation frequency.

The transcription factor analysis further supports the machine learning predictions. Not only is ATF7IP statistically identified by XGBoost as a novel candidate driver gene (64%), but it is also physically targeted by FOXA1, amplified and transcriptionally co-regulated with it. These three signals provide evidence for the dual-activation mechanism hypothesis.

However, this study has limitations. First, the thresholds used to assess statistical significance while analyzing each layer determine the results. Moreover, selecting only genes that are significant in at least two of the three omics layers excludes driver genes with only a single alteration, even if that alteration is major (**Figure 4**). Second, the definition of non-driver genes is based solely on exclusion from reference databases and introduces a serious bias as some of the excluded genes are certainly important driver of LUAD that have not yet been discovered or annotated as such. Furthermore, the lack of data on protein-protein interaction networks prevents the algorithm from understanding how genes interact with one another, which is essential in cancers. Also, the absence of epigenetic data, such as DNA methylation, prevents the identification of driver genes that are dysregulated exclusively by this specific mechanism. Finally, the results, so far, are only probabilities and without wet-lab experiments or validation on an independent external cohort, they remain hypothetical and not proven facts.

Conclusion

The first phase of the project aimed to identify a set of LUAD candidate driver genes by integrating three omics layers (transcriptome, mutations, and copy number alterations). The strict intersection approach proved to be too conservative, whereas the more flexible integration (Tier 2, 593 genes) combined with an XGBoost model produced a robust classification (MCC = 0.684, sensitivity = 0.923), identifying the canonical driver genes of LUAD while proposing ‘novel’ candidates, including ATM, which stands out due to its convergence across Random Forest and XGBoost models.

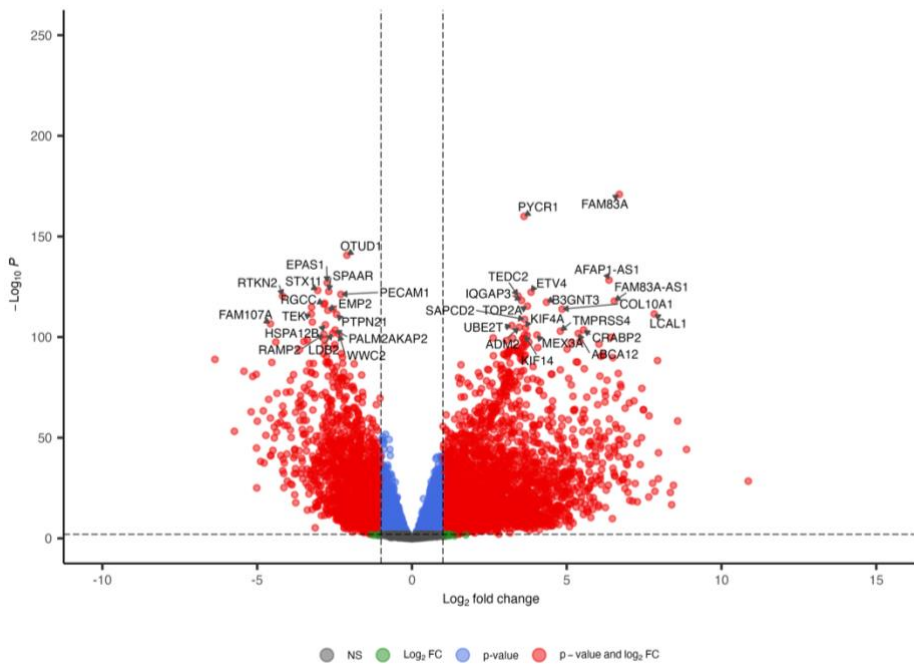
Phase 2, currently under development, aims to model sensitivity to anticancer drugs (ln(IC50)) by integrating the molecular structure of the drugs and the multi-omic profiles of the selected gene panel

(XGBoost probability $\geq 60\%$, ADME genes, and OncoKB genes). The ultimate goal will be to evaluate the clinical relevance of these predictions through direct inference on the TCGA-LUAD patient cohort and explainable AI tools.

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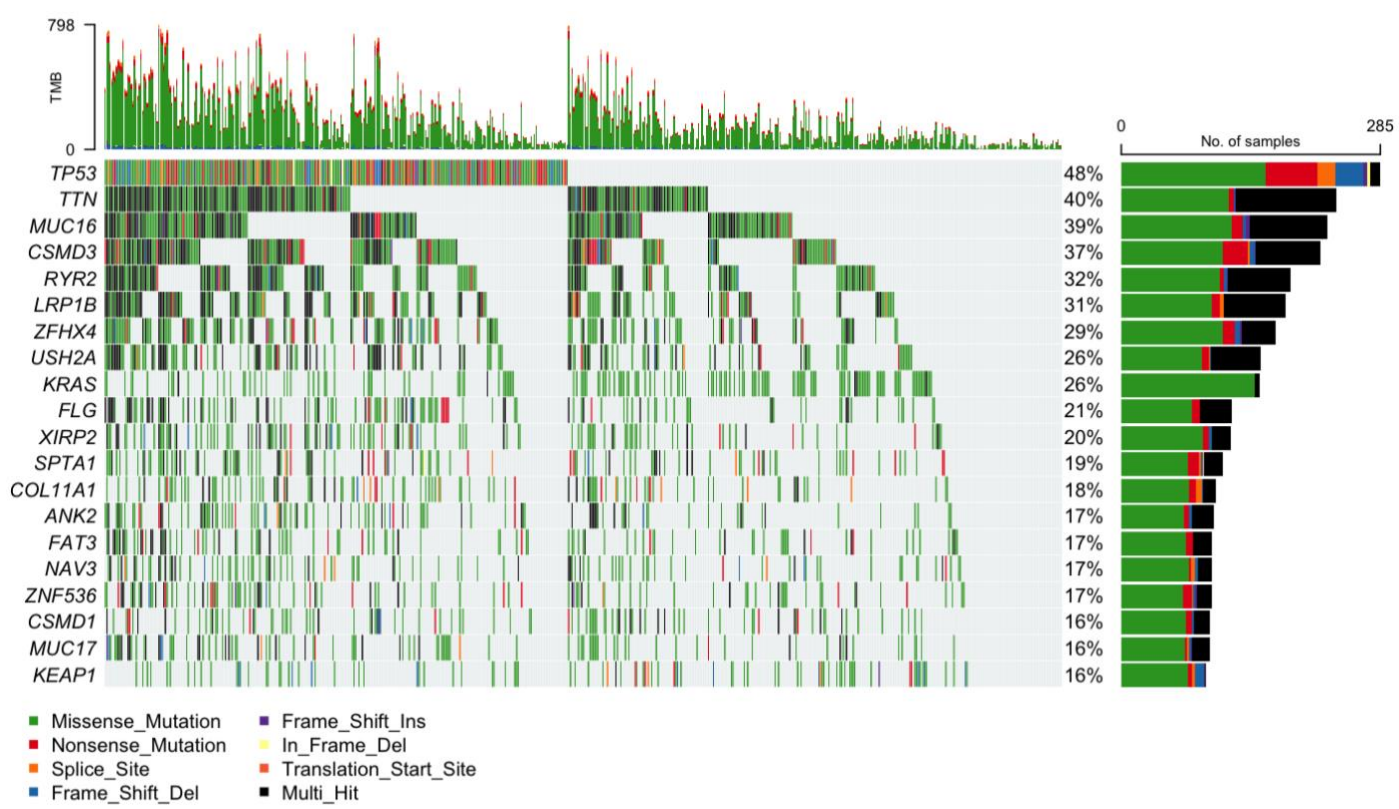
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Appendix



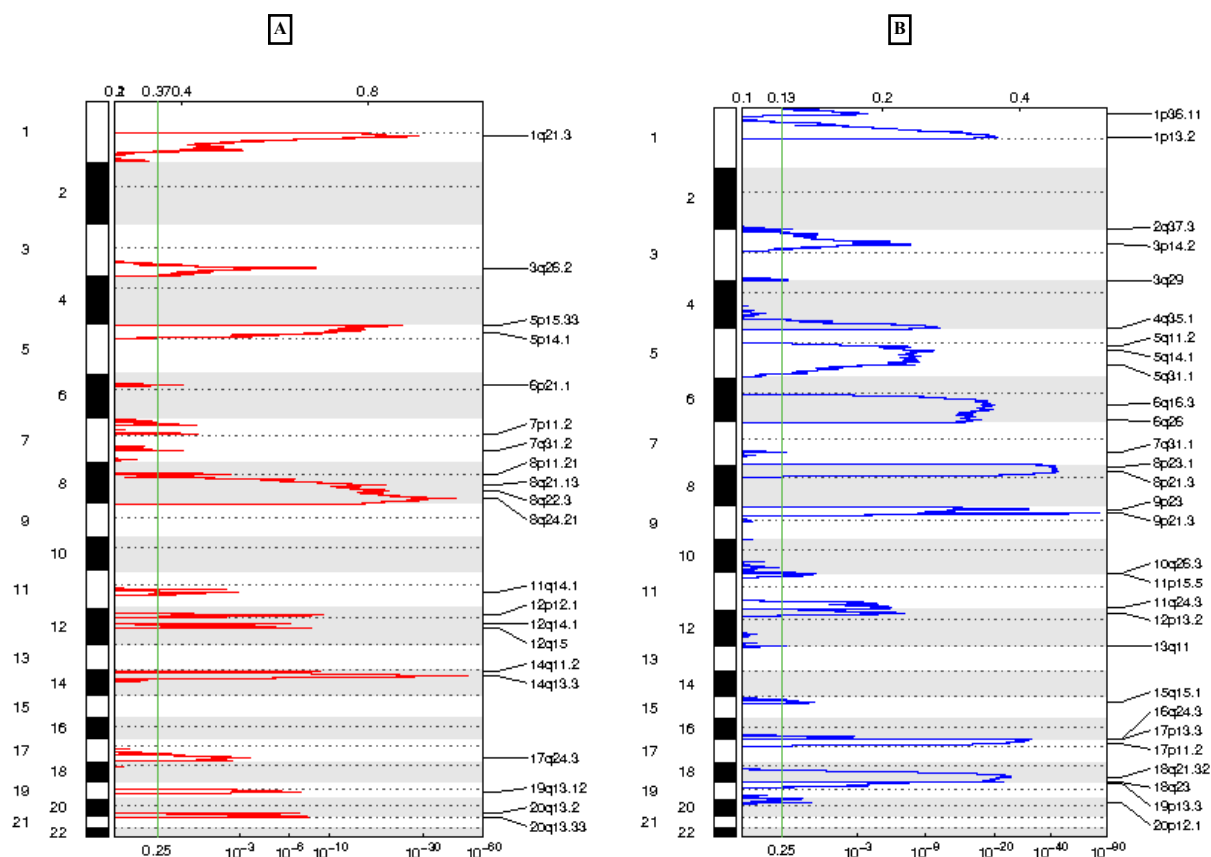
Appendix 1: Volcano plot of differential gene expression in TCGA-LUAD

Differential expression analysis of 20,739 genes between tumor and healthy tissues, adjusted for clinical covariates. The x-axis shows the magnitude of dysregulation ($\log_2FC$) and the y-axis shows statistical significance ($-\log_{10}$ adjusted p-value). Vertical dashed lines mark the $\pm 1 \log_2FC$ threshold. Red points indicate genes meeting both significance criteria ($padj < 0.01$ and $|\log_2FC| > 1$) and are therefore classified as differentially expressed genes (DEGs). The 20 most dysregulated genes are labeled by their gene symbol.



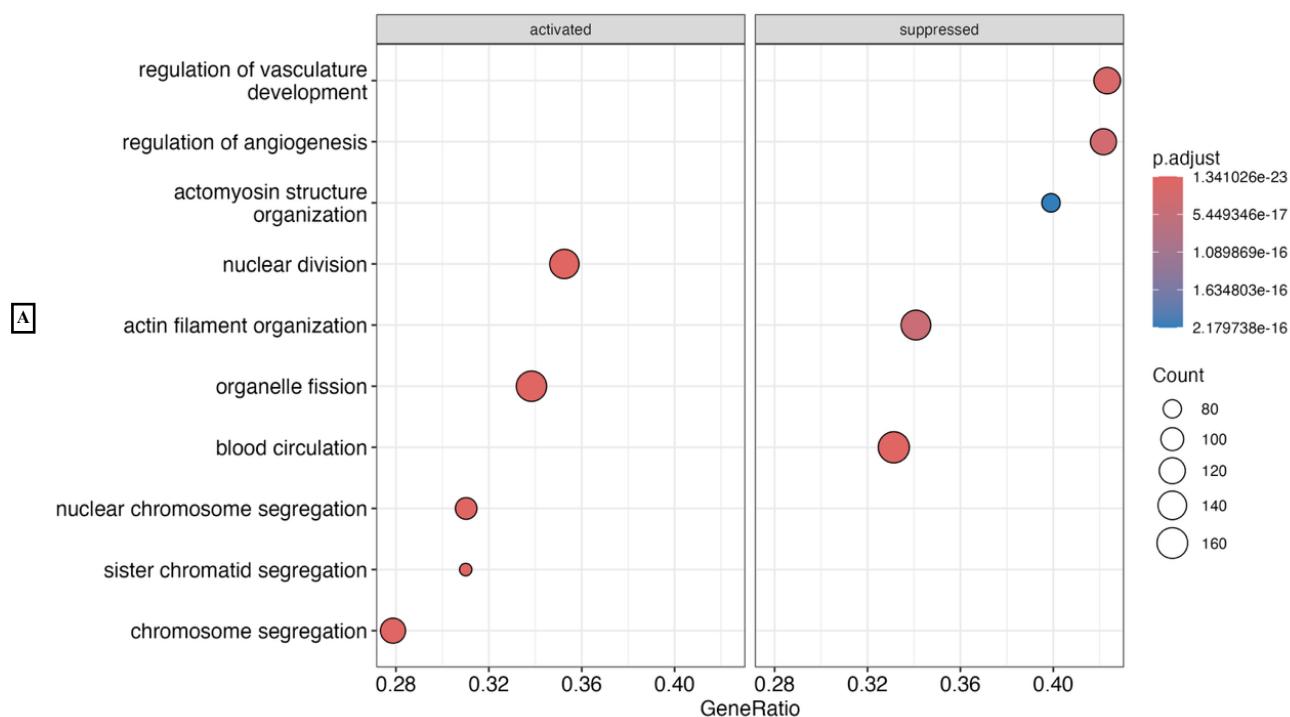
Appendix 2: Oncoplot of the 20 most frequently mutated genes in the TCGA-LUAD cohort

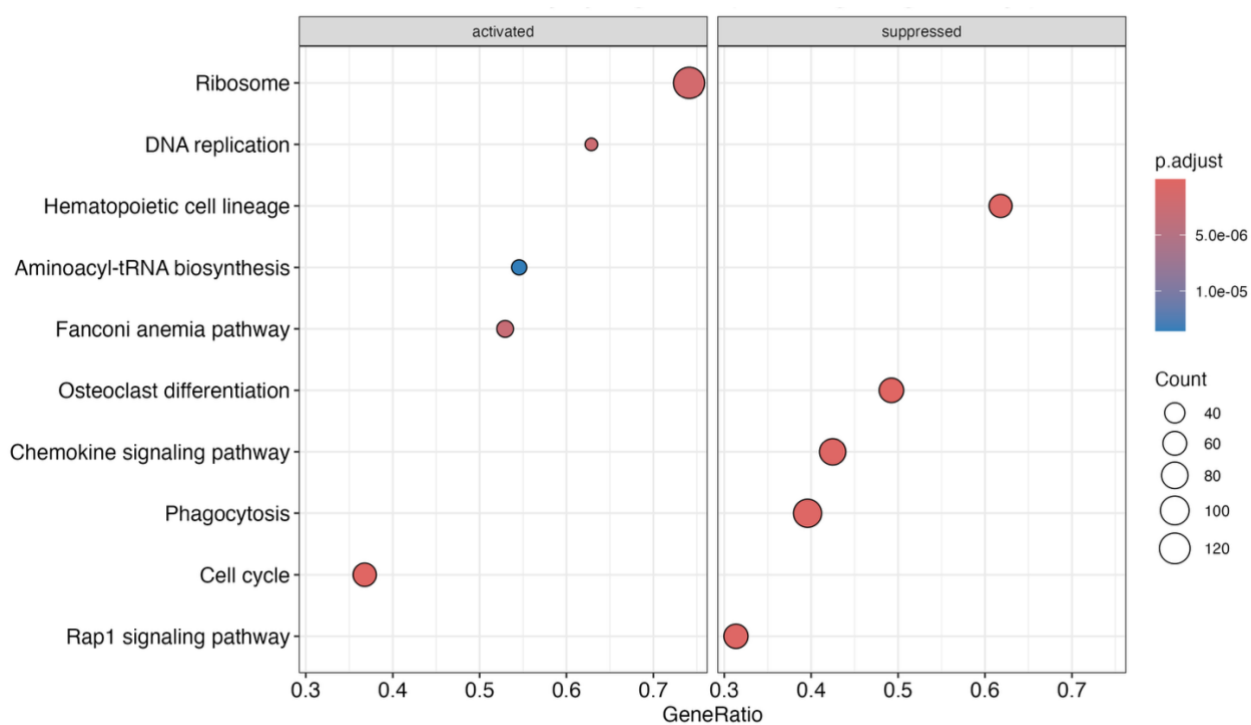
Each column represents one sample, and each row represents one gene, ranked by mutation frequency. Colored marks indicate the type of mutation observed, with black indicating multiple occurring alterations in the same sample (multi-hit). The top bar chart shows the total mutational burden (TMB) per sample.



Appendix 3: GISTIC2.0 significance plots for copy number amplifications (A) and deletions (B)

Each row corresponds to a chromosome. The top axis shows the significance score (q-value on logarithmic scale). Curves extending further right indicate stronger statistical significance. The green vertical line marks the default GISTIC significance threshold ($q < 0.25$) used to generate these plots. Cytobands reaching this threshold are labeled on the right.



B

Appendix 4: Top 5 dysregulated pathways per direction. GO results (A) and KEGG results (B)

GSEA results restricted to the five most significantly activated and suppressed pathways in each database. The x-axis shows the GeneRatio (proportion of leading-edge genes relative to pathway size), dot size indicates the number of core enrichment genes (Count), and color indicates the adjusted p-value.

Summary

Le cancer du poumon étant la première cause de mortalité par cancer dans le monde, s'intéresser aux gènes moteurs de cette maladie est primordial pour adapter les soins des patients. Ce projet a pour objectif d'identifier ces gènes, puis d'utiliser leurs caractéristiques pour prédire la réponse des patients à certains anticancéreux.

L'expression des gènes, les mutations et les altérations du nombre de copies de la cohorte TCGA-LUAD ont été utilisées. Chaque type de données été analysé séparément, puis les trois ont été combinés permettant d'obtenir un groupe de 593 gènes candidats. L'analyse d'enrichissement des voies de signalisation a révélé une activation des voies de prolifération et une inhibition des voies immunitaires et vasculaires (phénotype tumoral prolifératif).

Ces gènes ont ensuite été utilisés pour entraîner les modèles de Machine Learning. XGBoost s'est montré le plus performant, avec une bonne capacité à retrouver les vrais gènes moteurs du LUAD comme KRAS et TP53 (sensibilité de 0,923) tout en limitant les erreurs. Il a aussi proposé de 'nouveaux' gènes candidats pertinents, notamment ATM, retrouvé également par le modèle Random Forest. Les données ChIP-seq ont montré que ATF7IP, un 'nouveau' gène candidat, est physiquement lié au facteur de transcription FOXA1 et amplifié, renforçant l'hypothèse d'un éventuel mécanisme de double activation.

Ce travail constitue la base de la deuxième phase du projet, qui vise à prédire la sensibilité des patients aux anticancéreux à partir du profil des gènes identifiés.

Lung cancer being the leading cause of cancer-related deaths worldwide, understanding the genes that drive this disease is essential for tailoring patient care. The goal of this project is to identify these genes and then use their characteristics to predict patients drug response.

Gene expression, mutations, and copy number alteration from the TCGA-LUAD cohort were used. Each type of data was first analyzed, and then all three were combined to identify a set of 593 candidate genes. Pathway enrichment analysis revealed activation of proliferative pathways and suppression of immune/vascular pathways (proliferative tumor phenotype).

These genes were used to train the Machine Learning models. XGBoost proved to be the best performing model, demonstrating a strong ability to identify the true driver genes of LUAD, such as KRAS and TP53 (sensitivity of 0.923), while minimizing errors. But it also proposed 'novel' candidates driver genes, such as ATM, which was also identified by the Random Forest model. ChIP-seq data showed that one 'novel' candidate, ATF7IP, is physically bound by the transcription factor FOXA1 and amplified, supporting a possible dual-activation mechanism.

This work forms the basis for the second phase of the project, which aims to predict patients response to anticancer drugs based on the gene profiles identified.